TEXAS INSTRUMENTS / NATIONAL SEMICONDUCTOR
LM7805 3-Terminal 1.5A Positive Voltage Regulator

1. General Description
The LM7805 is a 3-terminal positive voltage regulator designed for local on-card regulation. It employs internal current-limiting, thermal-shutdown, and safe-area compensation.

Key Specifications:
- Output Voltage: 5.0 V DC (+/- 4% over temperature)
- Maximum Input Voltage: 35 V DC (Absolute Maximum)
- Recommended Input Voltage Range: 7.0 V to 25.0 V
- Dropout Voltage: 2.0 V typical at IOUT = 1.0 A
- Output Current: Up to 1.5 A with adequate heatsinking
- Quiescent Current: 5.0 mA typical (8.0 mA max)
- Ripple Rejection: 78 dB at 120 Hz

Pin Assignment (TO-220 Package):
- Pin 1: Input (Unregulated DC, minimum 7V)
- Pin 2: Ground
- Pin 3: Output (+5.0V Regulated)

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2. Thermal Considerations
Power dissipation is calculated as:
PD = (VIN - VOUT) * IOUT + (VIN * IQ)
For example, with VIN = 12 V, VOUT = 5 V, and IOUT = 500 mA:
PD = (12 - 5) * 0.5 A = 3.5 Watts.
Without a heatsink, the thermal resistance junction-to-ambient (RthetaJA) is 50 deg C/W.
Temperature rise = 3.5 W * 50 deg C/W = 175 deg C above ambient!
The IC will quickly reach thermal shutdown (150 deg C junction). A substantial heatsink is mandatory.

3. External Capacitors
- Input Capacitor: A 0.33 uF ceramic or tantalum capacitor is required if the regulator is located more than 2 inches from the power supply filter capacitor.
- Output Capacitor: An output capacitor of 0.1 uF ceramic is recommended to improve transient response and stability.